

The mixed-kernel Schatten estimate fails below $p = 2$

A counterexample inside the fractional-commutator class

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Status. Candidate counterexample, likely valid pending expert review.

1 Source question

Gerontogiannis and Mesland consider an Ahlfors d_f -regular compact metric measure space (X, d, μ) and, for $0 < 2\alpha < \beta \leq 1$, the commutator kernel

$$K_{\alpha,h}(x, y) = \frac{h(y) - h(x)}{d(x, y)^{d_f + 2\alpha}}, \quad h \in \text{Höl}_\beta(X, d). \quad (1)$$

In Remark 2.18 of [1], on PDF page 20, they ask whether for $1 < p < 2$ and $q = p/(p - 1)$ the estimate

$$\|K_{\alpha,h}\|_{\mathcal{S}_p} \leq \left(\int_X \left(\int_X |K_{\alpha,h}(x, y)|^q d\mu(x) \right)^{p/q} d\mu(y) \right)^{1/p} \quad (2)$$

continues to hold. For $p \geq 2$ this is the symmetric-kernel specialization of the Russo–Goffeng mixed-norm estimate [2]. An affirmative answer would improve the threshold in their Proposition 2.17.

At this point the role of $\ell \in \mathbb{N}$ becomes apparent, as it is used to speed-up the averaging process in (2.7) so that (2.11) converges to zero as n goes to infinity. For

$$\ell = \lceil d_f \beta^{-1} \rceil + 1$$

it holds that

$$0 < \beta - 2\alpha \leq \ell\beta - d_f - 2\alpha \leq \ell - d_f - 2\alpha - 1 + \beta.$$

Therefore, we have

$$\|K - F_n\| \lesssim \text{Hö}_\beta(h) \theta^{(\beta-2\alpha)n}$$

and by recalling that F_n has rank at most $\#\mathcal{D}_{\ell n} \lesssim \theta^{-d_f \ell n}$, the proof that $K \in \mathcal{S}_p(L^2(X, \mu))$ for every $p > d_f \ell(\beta - 2\alpha)^{-1}$ is immediate.

Finally, assuming $d_f(\beta - 2\alpha)^{-1} \geq 2$ yields sharper estimates. From [29, Theorem 2.3], when $p \geq 2$ and $q = p/(p-1)$ is its conjugate, it holds that

$$(2.12) \quad \|K\|_{\mathcal{S}_p} \leq \left(\int_X \left(\int_X |K(x, y)|^q d\mu(x) \right)^{\frac{p}{q}} d\mu(y) \right)^{\frac{1}{p}}.$$

In particular, for every $p > d_f(\beta - 2\alpha)^{-1}$ the right hand-side of (2.12) is less than or equal to a constant multiple of $\text{Hö}_\beta(h)$. $\square$

Remark 2.18. To our knowledge, it is an open problem whether inequality (2.12) holds for $1 < p < 2$. In case the answer to this problem is affirmative, then in Proposition 2.17 we can choose $\ell = 1$ even if $d_f(\beta - 2\alpha)^{-1} < 2$.

The proof of the next result is inspired by the proof of [21, Lemma 3.1], which is about commutators with Riesz potentials on Euclidean spaces. We note that Riesz potentials correspond to negative powers of the classical Laplacian.

Lemma 2.19. *Let $0 < 2\alpha < \beta \leq 1$ and $h \in \text{Hö}_\beta(X, d)$. Also, let $\mathcal{D}$ be a system of dyadic cubes as in (1.2) with expectation operators $\{E_n\}_{n \geq 0}$ as in (1.3). Then, for every $p > \max\{p(\alpha, \beta), 1\}$ with $p(\alpha, \beta)$ as in Proposition 2.17 and $\gamma = 2\alpha + d_f p^{-1}$, we have*

$$(2.13) \quad \sum_{n=0}^{\infty} \theta^{-\gamma p n} \|E_{n+1} h - E_n h\|_{L^p}^p \lesssim \|K_{\alpha, h}\|_{\mathcal{S}_p}^p.$$

Consequently, we have

$$(2.14) \quad \sum_{n=0}^{\infty} \theta^{-\gamma p n} \|E_n h - h\|_{L^p}^p \lesssim \|K_{\alpha, h}\|_{\mathcal{S}_p}^p.$$

Proof. The inequalities of the statement are well-defined since h and each $E_n h$ lie in $L^p(X, \mu)$. Also, from Proposition 2.17 we have that $\|K_{\alpha, h}\|_{\mathcal{S}_p} < \infty$. We first prove inequality (2.13). From property (v) of $\mathcal{D}$ and the Ahlfors d_f -regularity of μ we get

$$\begin{aligned} \sum_{n=0}^{\infty} \theta^{-\gamma p n} \|E_{n+1} h - E_n h\|_{L^p}^p &= \sum_{n=0}^{\infty} \theta^{-\gamma p n} \sum_{D \in \mathcal{D}_n} \int_D |E_{n+1} h(x) - E_n h(x)|^p d\mu(x) \\ &\lesssim \sum_{n=0}^{\infty} \theta^{-2\alpha p n} \sum_{D \in \mathcal{D}_n} \int_D |E_{n+1} h(x) - E_n h(x)|^p d\mu(x). \end{aligned}$$

Also, for every $n \geq 0$ and $D \in \mathcal{D}_n$ it holds that

$$(E_{n+1} h - E_n h)\chi_D = \sum_{u=1}^{\#\mathcal{C}(D)-1} \langle h, h_{D,u} \rangle_{L^2 h_{D,u}},$$

2 Proof intuition

The binary ultrametric Cantor space allows the infinite-dimensional kernel to be reduced exactly to four cylinders. Choose h to have two very close values on each of the two first-level halves. The larger fractional weight inside a half compensates for that small difference. As the ultrametric scale tends to zero, all six entries above the diagonal of the resulting 4×4 antisymmetric matrix tend to one.

That limiting matrix has singular values $\sqrt{2}+1$ and $\sqrt{2}-1$, each with multiplicity two, whereas every column has three unit entries. The resulting scalar inequality is strict in the wrong direction for every $1 < p < 2$. Continuity then gives a genuine example at a finite ultrametric scale. This directly addresses the restrictive reading in which the source's K must be the fractional commutator kernel (1).

3 Counterexample

Theorem 1. *For every $1 < p < 2$, there exist*

- *a compact Ahlfors-regular metric measure space (X, d, μ) ,*
- *parameters $\alpha = 1/8$ and $\beta = 1$, and*
- *a real locally constant (hence Lipschitz) function h ,*

such that $p > d_f/(\beta - 2\alpha)$ and the fractional commutator kernel (1) violates (2). In particular, the answer to Remark 2.18, with the displayed constant-one inequality as written, is negative throughout $1 < p < 2$.

Proof. Fix $p \in (1, 2)$ and write $q = p/(p-1)$. Let $X = \{0, 1\}^{\mathbb{N}}$ with Bernoulli measure assigning mass 2^{-k} to each cylinder of length k . For $0 < r < 1$, set

$$d_r(x, y) = r^{j(x, y)},$$

where $j(x, y) \geq 0$ is the index of the first coordinate at which x and y differ. This is the standard binary ultrametric, and μ is Ahlfors regular of dimension

$$d_f = \frac{\log 2}{\log(1/r)}.$$

Indeed, balls at the discrete radii r^k are cylinders of measure 2^{-k} ; the estimates between adjacent radii change only by fixed factors.

Set $\alpha = 1/8$ and $\beta = 1$. Denote the four length-two cylinders, in lexicographic order, by E_0, E_1, E_2, E_3 , and put

$$c = r^{-(d_f + 2\alpha)} = 2r^{-1/4}.$$

Define the locally constant real function h by the values

$$(h|_{E_0}, h|_{E_1}, h|_{E_2}, h|_{E_3}) = (0, c^{-1}, 1, 1 + c^{-1}). \quad (3)$$

This function is Lipschitz: it is constant within every length-two cylinder, and only finitely many positive distances occur between distinct such cylinders.

The kernel $K_{\alpha, h}$ vanishes on every $E_i \times E_i$. Between two distinct length-two cylinders it is constant. The distance between sibling cylinders is r , while the distance between cylinders in

different first-level halves is one. Consequently, on the orthonormal basis $e_j = 2\mathbf{1}_{E_j}$ of the four-dimensional cylinder-constant subspace, the integral operator has matrix $A_c/4$, where A_c is real antisymmetric and its upper-triangular entries are

$$(A_{01}, A_{02}, A_{03}, A_{12}, A_{13}, A_{23}) = (1, 1, 1 + c^{-1}, 1 - c^{-1}, 1, 1). \quad (4)$$

The operator annihilates the orthogonal complement of this subspace, because the kernel is constant in each variable on every cylinder pair. Thus its nonzero singular values are exactly those of $A_c/4$.

As $r \downarrow 0$, we have $c \rightarrow \infty$ and $A_c \rightarrow A$, where

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 \end{pmatrix}. \quad (5)$$

The singular values of A are

$$s_+ = \sqrt{2} + 1, \quad s_+ = \sqrt{2} + 1, \quad s_- = \sqrt{2} - 1, \quad s_- = \sqrt{2} - 1.$$

For completeness, A is skew-symmetric, its squared Frobenius norm is 12, and its Pfaffian is 1. Hence the two distinct squared singular values have sum 6 and product 1, giving $3 \pm 2\sqrt{2}$.

Every column of A has three entries of modulus one. Therefore, for the limiting four-block kernel, the p th power of the ratio of the left side of (2) to the right side is

$$R_p^p = \frac{(\sqrt{2} + 1)^p + (\sqrt{2} - 1)^p}{2 \cdot 3^{p-1}}. \quad (6)$$

We show that $R_p > 1$ whenever $1 < p < 2$. Put $a = 1 + \sqrt{2}$, so that $a^{-1} = \sqrt{2} - 1$, and define

$$F(t) = \log\left(\frac{a^t + a^{-t}}{2}\right) - (t - 1) \log 3.$$

Since $a^2 + a^{-2} = 6$, one has $F(2) = 0$. Moreover,

$$F'(t) = \log(a) \frac{a^t - a^{-t}}{a^t + a^{-t}} - \log 3 < \log(a) - \log 3 < 0.$$

Thus F is strictly decreasing and $F(p) > F(2) = 0$ for $p < 2$. Equation (6) gives $R_p > 1$.

Both sides of (2) are continuous functions of the entries of the finite matrix A_c . Hence, for all sufficiently large finite c (equivalently, all sufficiently small positive r), A_c still violates (2). Finally,

$$\frac{d_f}{\beta - 2\alpha} = \frac{4 \log 2}{3 \log(1/r)} \rightarrow 0 \quad (r \downarrow 0),$$

so the same choice of sufficiently small r also ensures $p > d_f/(\beta - 2\alpha)$. All hypotheses relevant to the proposed improved range are satisfied. $\square$

4 A sharp obstruction for arbitrary antisymmetric kernels

The preceding theorem refutes the constant-one estimate inside the special fractional family. The next observation explains why no generic antisymmetric-kernel argument can repair it by inserting a constant depending only on p .

Proposition 2. *Fix $1 < p < 2$. Among bounded real antisymmetric step kernels on the Ahlfors regular space $[0, 1]$, the ratio of the two sides of (2) is unbounded. More precisely, for every power of two n there is such a kernel with ratio exactly $n^{1/p-1/2}$.*

Proof. Let H_n be the Sylvester Hadamard matrix and $U_n = n^{-1/2}H_n$, so U_n is real orthogonal and all its entries have modulus $n^{-1/2}$. Set

$$B_n = \begin{pmatrix} 0 & U_n \\ -U_n^\top & 0 \end{pmatrix}.$$

Then B_n is a real antisymmetric orthogonal matrix of size $2n$. Partition $[0, 1]$ into $2n$ equal intervals and use B_n as the matrix of a step kernel. The integral operator has matrix $B_n/(2n)$ on normalized interval indicators, hence Schatten norm $(2n)^{1/p-1}$.

Each kernel column is nonzero on exactly n intervals, with modulus $n^{-1/2}$. Its mixed norm is

$$2^{-1/q}n^{-1/2}.$$

The quotient is therefore

$$\frac{(2n)^{1/p-1}}{2^{-1/q}n^{-1/2}} = n^{1/p-1/2} \longrightarrow \infty.$$

□

5 Scope, verification, and novelty

Theorem 1 disproves the exact inequality (2.12) with constant one, even when K is the source’s own fractional commutator kernel. It does *not* rule out an estimate with a factor depending on p (and possibly on geometric regularity constants), nor does it rule out proving the $\ell = 1$ improvement by a different argument. Proposition 2 shows only that such a repair cannot hold uniformly for all antisymmetric kernels; the rigid fractional-difference structure would have to be used.

The accompanying script checks the singular values, finite- c ratios, and the exact Hadamard family formulas. These checks are not part of the proof.

A bounded search on 17 August 2026 covered the current arXiv record for [1] (still v1), exact queries for Remark 2.18 and inequality (2.12), and fractional-commutator, ultrametric, and mixed Schatten-norm variants. No later resolution or this four-cylinder construction was found. The classical fact that kernel criteria below $p = 2$ are substantially more delicate, including Carleman-type obstructions, is reviewed by Delgado and Ruzhansky [3]; that general literature does not supply the special fractional-commutator example above. Novelty confidence is moderate pending specialist review.

Human review recommendation. Check the cylinder-distance convention and the resulting six entries in (4); the finite-rank reduction; and whether the authors intended an unspecified multiplicative constant rather than the displayed constant-one inequality. The last point affects the scope, not the validity, of the stated counterexample.

References

- [1] D. M. Gerontogiannis and B. Mesland, *Ideal quantum metrics from fractional Laplacians*, arXiv:2502.04187v1 (2025).

- [2] M. Goffeng, *Analytic formulas for the topological degree of non-smooth mappings: the odd-dimensional case*, Adv. Math. 231 (2012), 357–377.
- [3] J. Delgado and M. Ruzhansky, *Schatten–von Neumann classes of integral operators*, J. Math. Pures Appl. 154 (2021), 1–29.